

III B.TECH	FULL STACK DEVELOPMENT -2	L	T	P	C
I SEMESTER		0	1	2	2

Prerequisites: Basics of Java

Course Objectives

- Understand the architecture and workflow of full-stack web applications.
- Define and handle routes, including route and query parameters.
- Implement and handle event listeners in React components.
- Create a basic HTTP server using Node's core modules.
- Prepare for industry-readiness through hands-on MongoDB certification & deployment skills.

Course Outcomes:

On successful completion of the course, students will be able to:

1. Structure and implement HTML/CSS.
2. Apply intermediate and advanced web development practices.
3. Implement basic JavaScript.
4. Create visualizations in accordance with UI/UX theories.
5. Develop a fully functioning website and deploy on a web server.

List of Experiments:

1: ExpressJS – Routing, HTTP Methods, Middleware.

- a. Write a program to define a route, Handling Routes, Route Parameters, Query Parameters and URL building.
- b. Write a program to accept data, retrieve data and delete a specified resource using http methods.
- c. Write a program to show the working of middleware.

2: ExpressJS – Templating, Form Data

- a. Write a program using templating engine.
- b. Write a program to work with form data.

3: ExpressJS – Cookies, Sessions, Authentication

- a. Write a program for session management using cookies and sessions.
- b. Write a program for user authentication.

4: ExpressJS – Database, RESTful APIs

- a. Write a program to connect MongoDB database using Mongoose and perform CRUD operations.
- b. Write a program to develop a single page application using RESTful APIs.

5: ReactJS – Render HTML, JSX, Components – function & Class

- a. Write a program to render HTML to a web page.
- b. Write a program for writing markup with JSX.
- c. Write a program for creating and nesting components (function and class).

6: ReactJS – Props and States, Styles, Respond to Events

- a. Write a program to work with props and states.
- b. Write a program to add styles (CSS & Sass Styling) and display data.
- c. Write a program for responding to events.

7: ReactJS – Conditional Rendering, Rendering Lists, React Forms

- a. Write a program for conditional rendering.
- b. Write a program for rendering lists.
- c. Write a program for working with different form fields using react forms.

8: ReactJS – React Router, Updating the Screen

- a. Write a program for routing to different pages using react router.
- b. Write a program for updating the screen.

9: ReactJS – Hooks, Sharing data between Components

- a. Write a program to understand the importance of using hooks.
- b. Write a program for sharing data between components.

10: ReactJS Applications – To-do list and Quiz

- a. Design to-do list application.

11: Node.js

- a. Write a program to show the workflow of JavaScript code executable by creating web server in Node.js.
- b. Write a program to transfer data over http protocol using http module.
- c. Create a text file src.txt and add the following content to it. (HTML, CSS, Javascript, Typescript, MongoDB, Express.js, React.js, Node.js)
- d. Write a program to parse an URL using URL module.
- e. Write a program to create an user-defined module and show the workflow of Modularization of application using Node.js

12: Typescript

- a. Write a program to understand simple and special types.
- b. Write a program to understand function parameter and return types.

- c. Write a program to show the importance with Arrow function. Use optional, default and REST parameters.
- d. Write a program to understand the working of typescript with class, constructor, properties, methods and access specifiers.
- e. Write a program to understand the working of namespaces and modules.
- f. Write a program to understand generics with variables, functions and constraints.

13: MongoDB – Installation, Configuration, CRUD operations

- a. Install MongoDB and configure ATLAS
- b. Write MongoDB queries to perform CRUD operations on document using insert(), find(), update(), remove()

14: MongoDB – Databases, Collections and Records

- a. Write MongoDB queries to Create and drop databases and collections.
- b. Write MongoDB queries to work with records using find(), limit(), sort(), createIndex(), aggregate().

15: Augmented Programs: (Any 2 must be completed)

- a. Write a CSS program, to apply 2D and 3D transformations in a web page.
- b. Design a web page with new features of HTML5 and CSS3.
- c. Design a to-do list application using Javascript.
- d. Design a to-do list application using NodeJS and ExpressJS.
- e. Design a Quiz app using ReactJS.
- f. Complete the MongoDB certification from MongoDB University website.

Text Books:

1. Programming the World Wide Web, 7th Edition, Robert W Sebesta, Pearson, 2013.
2. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasan Subramanian, 2nd edition, APress, O'Reilly.

Reference Books:

1. ExpressJS - <https://www.tutorialspoint.com/expressjs>
2. ReactJS - <https://www.w3schools.com/REACT> (and) <https://react.dev/learn#>
3. MongoDB - <https://learn.mongodb.com/learning-paths/introduction-to-mongodb>

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